

The Problem

Repo- <https://github.com/MangosteenMadness/Biohack>

The Challenge

Participants will develop MCP tools, agents, and workflows that extend the capabilities of AI scientist systems like K Pro.

How to submit

- **Team size:** min =2, max = 5
- **Submit your creations by 4pm Sunday July 12th using this project submission template**
- **Minimally**
 - Demo video <5min (mp4, mov only)
- **Preferred**
 - Demonstration of a working system / service
 - Open github repo
 - Pitch deck or 1 pager (Google Docs / Google Slides OR Microsoft Word / Powerpoint only)

Research is Hard

Ask question- Has this been asked before- Internally first and externally second

One sentence Pitch-

The one sentence

An internal research-memory layer for an AI-scientist platform (K Pro): before a researcher runs a hypothesis, it checks — **internally first** — whether someone in their own org already explored it (including *abandoned* attempts), then whether the published world has. External novelty is commoditized (FutureHouse's Owl does it). **Internal, in-flight memory is the wedge nobody owns.**

her asks a survival-stratification question on TCGA LUAD (lung adeno). Agent runs it, gets a real result (e.g., "high [signature] → worse OS, HR ~1.8, n=230"). Agent writes it to the graph with provenance (who, when, data, effect) and a **calibrated** novelty note ("no prior internal work; published record shows related work in X").

Session 2 — the save. A *different* researcher (new session) asks a related question — e.g., the same signature in LUSC (lung squamous). Instead of cold-starting, the agent responds:

"Before you run this — someone in your org tested a version of this in Session 1 [here's who, here's what they found]. Separately, a teammate **abandoned** a related approach because [reason]. Published record adds [context]. The genuinely open part is [X]."

The punchline: internal recall + an abandoned-attempt surfaced = something Owl and the consulting knowledge graphs structurally cannot do, because both only see the *published* record where failures never appear.

Requirements

P0 — must work or there's no demo

1. **Graph store** with findings as nodes: {*disease, hypothesis_text, signature, effect, n, status, author, timestamp, provenance*}. Status ∈ {confirmed, in-progress, **abandoned**}.
2. **write_finding** — append-only, provenance-tagged.
3. **recall_findings** — given a new question, return semantically-related prior findings (LLM-matched).
4. **One real analysis** — TCGA slice → survival stratification → produces a finding object. Thin and real.
5. **The two-session flow** runs end-to-end at least once.

P1 — makes it win (add once P0 is solid)

6. **check_duplication** presenting **internal first, external second**.
7. **External literature check** — PubMed/Europe PMC API call (free), with cached fallback.
8. **Abandoned-result surfacing** as a first-class result type. *This is the differentiator — pre-seed 1–2 abandoned findings with reasons.*
9. **Calibrated language** everywhere: "no prior work found in [sources]," never "this is novel."

P2 — only if hours remain

10. Cross-indication recall shown as a graph visual.
11. A "novelty verdict" badge attached to each stored finding.

Technical architecture (lightest path)

- **MCP server (Python + FastAPI).** The deliverable *is* an MCP server — this reinforces the "add-on to K Pro, consumed via MCP" thesis. Tools: `run_analysis`, `write_finding`, `recall_findings`, `check_duplication`.
- **Graph store:** SQLite + a thin relational schema, or `networkx` in-memory persisted to JSON. **Do not stand up Neo4j** — setup cost isn't worth it in 13h.
- **Semantic matching:** Claude API — "are these two hypotheses the same question?" No embedding infra needed.
- **External check:** Europe PMC REST (no key) or PubMed E-utilities. Cache results so the demo can't be broken by a flaky network.
- **Data:** pre-stage one indication's **clinical + bulk RNA** before 12:00 if possible. cBioPortal API is the fastest way to pull TCGA slices. You need enough to run one real Kaplan-Meier/Cox, not all 10K patients.
- **Demo surface:** cleanest option — connect the MCP server to **Claude Desktop** and drive it live (the MCP host stands in for K Pro). **Record a backup video** of the full flow tonight — do not trust demo-gods at 16:00. Fallback: a 1-page thin chat UI (Next.js) you fully control.

Roles

Person	Owns
Biologist 1	Locks the disease + the ONE hypothesis. Picks the signature. Writes the biological demo script.
Biologist 2	Seeds the graph with realistic prior findings — incl. 1–2 abandoned ones with reasons. Live-validates that recalled matches are credible. On-stage "this is a real match" voice.
Builder 1	Data + analysis pipeline: TCGA slice → survival stratification → finding object. <code>run_analysis</code> tool.
Builder 2 (stronger)	The moat: graph store + <code>write_finding/recall_findings/check_duplication</code> + LLM semantic match + external lit check.

Product/Pitch (Yousuf) Agent orchestration glue, demo surface, scope enforcement, deck, narrative, Owkin positioning.

The demo (write everything toward this)

Session 1 — a finding is born. Researcher asks a survival-stratification question on TCGA LUAD (lung adeno). Agent runs it, gets a real result (e.g., "high [signature] → worse OS, HR ~1.8, n=230"). Agent writes it to the graph with provenance (who, when, data, effect) and a **calibrated** novelty note ("no prior internal work; published record shows related work in X").

Session 2 — the save. A *different* researcher (new session) asks a related question — e.g., the same signature in LUSC (lung squamous). Instead of cold-starting, the agent responds:

"Before you run this — someone in your org tested a version of this in Session 1 [here's who, here's what they found]. Separately, a teammate **abandoned** a related approach because [reason]. Published record adds [context]. The genuinely open part is [X]."

"Benchling Lab Data -

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We use LLM wiki style. Log and content so

Demo

Demo Script

Runtime: ~2.5 min live demo inside a ~5 min pitch. **Proof we're delivering:** memory that persists *across sessions* and catches duplication the *published* record can't see. **Primary hypothesis:** cytotoxic/immune signature (CD8A, GZMB, PRF1, GZMK) vs overall survival in LUAD. **Documented fallback:** STK11 mutation status vs OS in LUAD (swap nouns, same arc) — use if the immune split doesn't separate cleanly.

Calibration rule (non-negotiable): the agent NEVER says "this is novel." It says "no prior *internal* work found" and "published record shows related work."
Overclaiming loses the room.

Pre-demo state (loaded before we present)

Builder 2 pre-seeds the graph with **one abandoned finding** (see Appendix A). The graph is otherwise empty of the Session-1 finding — that gets written **live** so the audience watches the memory grow.

Number slots `{{ }}` are filled by Builder 1 from the real cBioPortal/TCGA LUAD run. Lock these by the 19:00 checkpoint tonight.

BEAT 0 — Cold open (Biologist 1, 20s, before touching the tool)

"Last year I spent three weeks building toward a hypothesis a colleague two floors up had already killed. It was never published — negative results never are — so no literature tool on earth could have warned me. That lost time is the most common, most invisible waste in research. Watch what happens when the platform *remembers*."

[Hand to presenter driving the tool.]

BEAT 1 — Session 1: a finding is born (Builder drives, 45s)

Presenter types:

In lung adenocarcinoma, is high cytotoxic T-cell infiltration associated with better overall survival?

Agent responds (runs analysis, then writes to graph):

Ran a high-vs-low split on a cytotoxic signature (CD8A, GZMB, PRF1, GZMK) across TCGA LUAD.

Result: high-infiltration patients ($n=\{N_HIGH\}$) showed better overall survival than low ($n=\{N_LOW\}$). HR $\{HR\}$ (95% CI $\{CI\}$), log-rank $p=\{P\}$. Median OS $\{MEDIAN_HIGH\}$ vs $\{MEDIAN_LOW\}$ months.

Novelty check: no prior *internal* work found in this workspace. Published record shows related work on immune infiltration and NSCLC prognosis — this is consistent with, not new to, the field.

Written to memory as finding #F-118 — attributed to *Researcher A*, $\{date\}$, with the effect size, cohort, and method. Anyone who asks a related question will now see this first.

Presenter, one line: "It didn't just answer — it kept the finding, with who asked it and what they found."

BEAT 2 — the persistence proof + the save (Builder + Biologist 2, 60s)

⚠ CRITICAL STAGE DIRECTION: Presenter **visibly starts a fresh session** — new chat / restart the client — and says it out loud: *"New session. Different researcher. Nothing in context. If this were just chat history, it would have no idea what just happened."* This is the credibility crux — it proves real persistence, not conversation memory.

Presenter types (deliberately different wording — same underlying question):

Does immune infiltration predict prognosis in lung cancer?

Agent responds:

Before you run this — a related question has already been explored in your org.

Internal match (finding #F-118): *Researcher A* tested whether a cytotoxic T-cell signature predicts overall survival in LUAD and found HR $\{\{HR\}\}$, $p=\{\{P\}\}$. Your question ("immune infiltration / prognosis / lung cancer") is the same hypothesis in different terms. Running it would largely repeat their work.

■ **Related abandoned attempt (finding #F-093):** a colleague tried to turn this immune signature into a *stage-independent* prognostic model in Q2 and **abandoned it** — the effect collapsed after adjusting for tumor stage, i.e. the signal was largely tracking stage, not independent biology. Their note: "*do not re-run without a multivariate, stage-adjusted design.*"

Published record: consistent with both. No *internal* work found on the stage-adjusted question since it was abandoned — that remains genuinely open.

Biologist 2, stepping in to validate — this is the money line: "That's a real match. Those two questions are the same biology in different words — a keyword search would have missed it. And that abandoned result? That's exactly the kind of thing that dies in someone's notebook. It just saved this researcher a week *and* pointed them at the one version of the question still worth doing."

BEAT 3 — land it (Yousuf, 15s, no tool)

"Every tool out there checks the *published* world — that's solved. Cairn is the only layer that checks your *own org's* in-flight and abandoned work, because it rides on the platform where the work actually happens. External novelty tells you if it's new to the world. We tell you if it's new to *you* — and point you at the colleague who already knows."

Appendix A — Seed data (Biologist 2 loads before demo)

Finding #F-093 — the abandoned attempt (MUST be pre-loaded):

Field	Value
disease	LUAD
hypothesis_text	"Immune/cytotoxic signature as a stage-independent prognostic model"

signature	CD8A, GZMB, PRF1, GZMK
status	abandoned
reason	"Effect collapsed after stage adjustment — signal tracked tumor stage, not independent biology"
note	"Do not re-run without multivariate, stage-adjusted design"
author	Researcher C
timestamp	~Q2 (any date before demo)

*Optional second seed (only if it strengthens, not clutters): one unrelated **confirmed** finding in a different indication, so the graph visibly contains more than just the demo path.*

Finding #F-118 (Session 1) is NOT pre-seeded — it is written live in Beat 1.

Appendix B — Number-fill checklist (Builder 1, lock by 19:00)

Fill from the real TCGA LUAD run so every `{{ }}` in Beats 1–2 is a true number:

- `{{N_HIGH}}` / `{{N_LOW}}` — cohort sizes of the high/low split
- `{{HR}}` + `{{CI}}` — hazard ratio and confidence interval
- `{{P}}` — log-rank p-value
- `{{MEDIAN_HIGH}}` / `{{MEDIAN_LOW}}` — median OS in months
- `{{date}}` — Session-1 timestamp

If immune signature doesn't separate cleanly: switch to STK11-mutant vs wild-type OS. Rephrase Beat 1 → "Is STK11 mutation associated with worse survival in LUAD?" and Beat 2 → "Does LKB1 loss affect lung cancer outcomes?" (same gene, different name — a *bonus* semantic-match demo). Update Appendix A's abandoned finding to a mutation-based framing.

Roles

Question Continuous Chat - Hypothesis Formed —Save Wiki - Store in Graph

Question - Retrieve - Output - Identical Problem Soling

Question- Retreive - Output - Leverage Previous abandoned stury

UX - Question Data chat -Yousuf

Save chat Question to Wiki - Mahdi

Graph Database and Schema- Aufstin

MCP and Retrieval -

Data - 3 Questions- Roopal

Tab 4

